

Sneha Sengupta

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Education

Duke University

MEng in Biomedical Engineering

2026-2027

BSE in Biomedical Engineering

2022-2026

Skills

- **Programming Languages:** Python, MATLAB-Simulink, C++
- **Softwares:** SolidWorks, Onshape, COMSOL Multiphysics, KiCad, Arduino
- **Wet Lab Skills:** Microscopy, Cell Passaging, Cell Seeding, Cell Culture Maintenance, Quantitative PCR, Collagen Gels, Protein Isolation, Western Blots, Viral Transfection, Gibson Assembly

Experience

Medical Affairs Intern - Asensus Surgical (a KARL STORZ Company)

Summer 2026

- Applied machine learning and computer vision feature extraction techniques to analyze robotic surgical telemetry data, generating clinically meaningful insights for surgeon performance and surgical quality.
- Supported the medical affairs team by translating data findings into actionable recommendations to enhance training effectiveness, decrease robotic-assisted surgery learning curves, and surgeon experience with the robotic system.

Pediatric Nephrology Research Laboratory

Spring 2024-Present

- Contributing to manuscripts, data analysis for biomedical datasets, translational research involving immunotherapy pathways & transplant rejection mechanisms analysis, reinforcing best practices for scientific communication and lab SOPs.

CRISP Intern - Duke School of Medicine

Fall 2025-Spring 2026

- Designed and executed a retrospective cohort study of pediatric kidney transplant patients ages 1-23 who were patients at Duke Medical Center between 2016 to 2025 to examine the clinical outcomes for patients treated with IVIG.
- Processed immunotherapy parameters and patient records to determine significance in trends and outcomes in regimens.

Medical Assistant Volunteer- Women's First OBGYN

Summer 2023

- Supported clinical staff by providing patient education, counseling resources, medical care, insurance validation and verification, and administrative assistance in a fast-paced outpatient healthcare setting.

Projects

Machine Learning

Breast Cancer Classification-Machine Learning

- Applied machine learning techniques (logistic regression, SVMs, neural networks) on the UCI Breast Cancer dataset to classify tumors as benign or malignant. Optimized parameters and evaluated models using accuracy and F1 score. Achieved > 95% accuracy with the tuned SVM and neural network models.

Electromechanical Integration

Lightbox

- Designed and built an Arduino-based "One Shot Blinking Light" system, developing custom firmware, PCB schematics, and a 3D-printed enclosure that integrated LEDs, MOSFET drivers, sensors, and user controls within 99% accuracy.

EMG-Controlled Prosthetic Hand

- Designed a prosthetic hand that captures EMG signals from forearm muscles, processes them through amplification and filtering, and drives motors via a microcontroller to produce motions in a 3D-printed hand.

Electrocardiogram Biopotential Amplifier

- Built and analyzed a multistage ECG amplifier circuit incorporating an instrumentation amplifier, high-pass and low-pass filters and an isolation amplifier for patient safety and noise rejection.

Baby Incubator Control System

- Designed & implemented a closed-loop temperature control system using a thermistor, relay, and Arduino microcontroller to maintain neonatal incubator temperature with sensor calibration, hysteresis, and OLED user interface for autonomous use.

Molecular & Cellular Design & Development

PEGylated Thrombomodulin Coatings for Decellularized Vascular Grafts

- Engineered a bidirectional PEGylated thrombomodulin surface coating for decellularized vascular grafts to improve biocompatibility, reduce thrombosis, and enhance endothelialization.

Low-Cost Microneedle Hydrogel Continuous Glucose Monitoring System

- Designed and created a low-cost, power-free continuous glucose monitoring system using microneedle hydrogel biosensors with GOx- and PBA-based chemistries to provide colorimetric readouts of hypo- and hyperglycemia.

AAV-TREM2 Gene Therapy Vector Design

- Engineered and conducted plasmid design of an adeno-associated virus (AAV-DJ) vector with a CAG promoter to over express the TREM2 gene in microglia to enhance amyloid- β clearance in Alzheimer's disease models in a wet lab setting.

Targeted In Vivo CCR5 Gene Editing for HIV Immunization and Therapy

- Designed a targeted gene therapy strategy using HIV1 pseudotyped lentiviral vector to deliver CRISPR-Cas9 for CCR5 knockout in CD4+ T cells to minimize lifelong treatment burden.

Transport Phenomena Dynamics & Kinetics

Modeling Radioimmunotherapy of Metastatic Melanoma

- Developed a mathematical model simulating antibody distribution & binding in tumor tissue across antigen concentrations.
- Analyzed and interpreted spatial profiles of antibody-antigen complex formation to identify diffusion limitations and binding site barriers to optimize treatment outcomes of therapies using COMSOL.

CFD Modeling of Variable-Height Flow Chamber for Cell Stress Analysis

- Simulated fluid flow in a variable-height chamber using 2D/3D models in COMSOL to predict shear stress on adhered cells.
- Validated analytical shear stress equations through numerical comparison and visualized fluid acceleration across chamber length given simulation constraints and prompts.

Targeted Immunomodulation of Transplants Using MMF-Nanoparticle Delivery

- Investigated localized drug delivery strategies to suppress immune rejection and response in organ transplantation recipients post transplant using MMF-loaded PEG-PLGA nanoparticles.
- Analyzed molecular immune pathways (IL-6, TNF α , IFN γ , PD-L1, CXCL9/10) contributing to graft inflammation and chronic rejection and proposed "smart" nanoparticle system to regulate immunosuppression from local inflammatory signals.

Pharmacokinetics of Drug Treatments for Treating Chicken Pox

- Developed and designed a comprehensive mathematical model that simulates the pharmacokinetics of acyclovir cream and interferon alpha therapy within the immune system of T-cells in response to chickenpox, integrating experimental parameters.

Toxicokinetics of BPA on the Human Body

- Developed and designed a comprehensive mathematical model that simulates the toxicokinetics of BPA within the endocrine system, capturing complex interactions between hormones and metabolic processes.
- Designed algorithms to model BPA absorption, distribution, metabolism, and excretion, ensuring accurate representation of its impact on human endocrine function integrating experimental parameters.

Acoustofluidics for Plasmapheresis

- Designing and developing a prototype device for plasmapheresis, creating detailed 3D models of microfluidic channels and integrating acoustic transducers to optimize fluid separation for a functional prototype that identifies immune sensitivity levels.

Presentations & Publications

Sengupta, Sneha, et al., "Maintenance IVIG Therapy Improves Clinical Outcomes in Adherent Pediatric Kidney Transplant Recipients" Proceedings of the *American Transplant Congress (ATC) Conference*, Boston, MA, June 2026

Sengupta, Sneha, et al., "Acute and Chronic Kidney Transplant Rejection in Adolescents: Causes and Treatment" Proceedings of the *Adolescents with Chronic Kidney Disease, Second Edition - Chapter 9*, 2026

Leadership

Orientation Program Director

- Co-designed a band camp schedule with Duke New Student and Family Programs and the Duke Marching Band director. Handled and oversaw logistics of travel, food, activities, and schedule planning for orientation week.

Progress Period

- Co-President, previously Vice President, Social Media Co-Chair, Education Committee Head (23-24), overseeing and managing Duke University's chapter of PERIOD. a global youth fueled non-profit which strives to eradicate period poverty and stigma through advocacy, education, and service.

Penny Pilgrim George Women's Leadership Initiative

- Volunteer partner with local nonprofit, Keep Durham Beautiful to create a "Wastewise Event Guide" as a resource for hosting sustainable events as part of the City of Durham Strategic Plan.

Duke University Marching & Pep Band

- Section & orientation leader enhancing performance quality through focus of musicality & mentorship of fellow students.

Duke On Tap

- Choreographer and Co-President for a tap dancing team. Responsible for coordinating rehearsals and performances, teaching choreography, and forming a creative and supportive environment.

Other Experience

Service Desk Worker- Duke University Library

2023-Present

Retail Sales Associate-Duke University Stores

2022-2023